CEITEC

MUNI



C9555: A survival guide to simulations

Week 4

Denys Biriukov – Oct. 19th, 2023

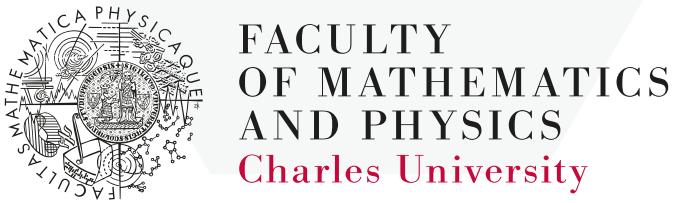
Acknowledgments

This part of the course is inspired and partially based on:

- Teaching materials of “Spring School in Computational Chemistry 2023” taken place in Finland and prepared by Dr. Hector Martinez-Seara and Dr. Matti Javanainen.
- Teaching course about molecular dynamics simulations at Charles University by prof. Lukasz Cwiklik.







More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

Get ready for practical session

- Wolf computers with GPU (more comfortable to work on)
  - wolf04 (not really good)
  - wolf05
  - wolf07
  - wolf08
  - wolf09
  - wolf12
  - wolf18
  - wolf19

Get ready for practical session

- A bit of things to be done:
  - `mv ~/survival_guide_simulations_c9555/week03/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- We will also check the disk usage together.
- We may transfer some of your data to MetaCentrum.

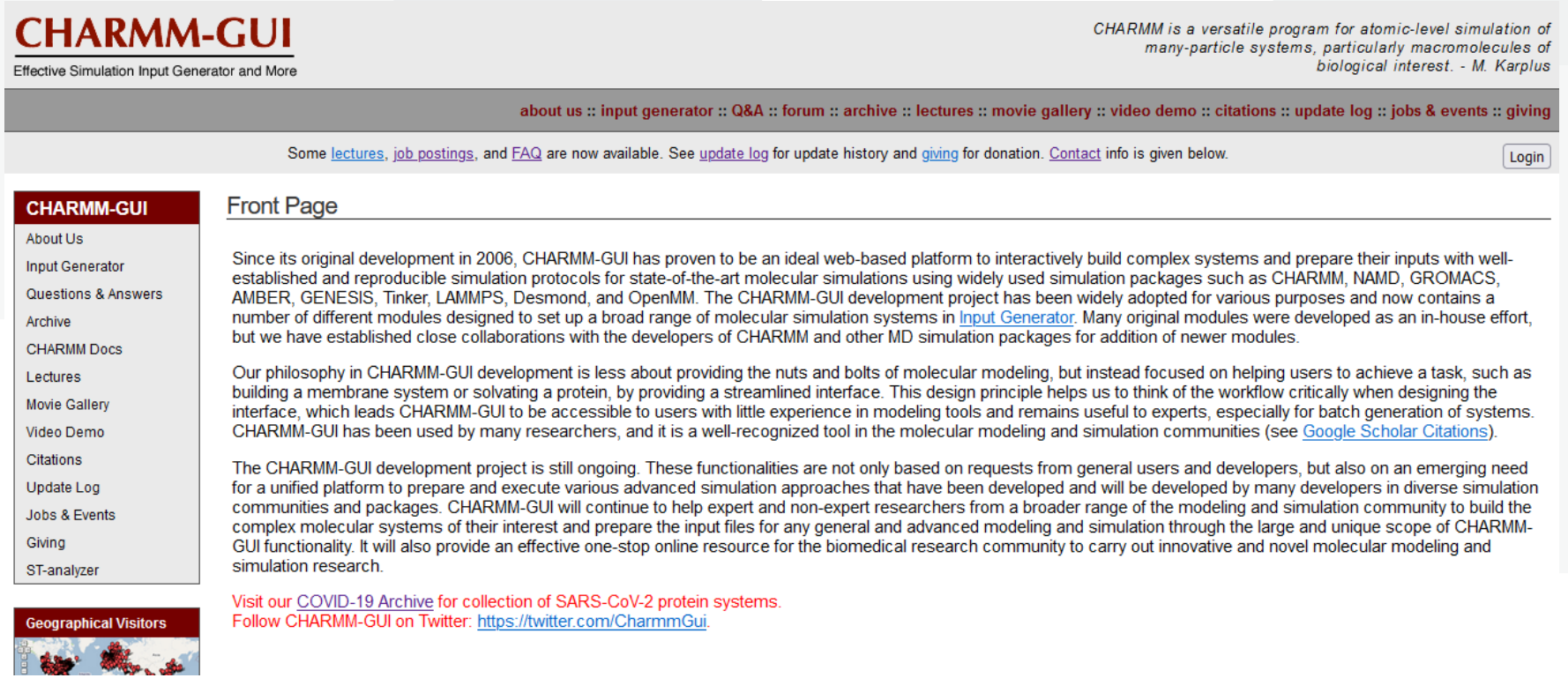
- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week04/`
- `bash ../infinity/infinity-temp.sh`
- `bash ../conda/conda-activate.sh`
- `bash ../conda/jupyter-start.sh`

CHARMM-GUI registration

- Necessary for the next week.
- https://www.charmm-gui.org/



What is today?

- Week 4:
  Ions, Molecules, and Their Interactions
- We are done with solvents only, time to put something in.
- We have a lot of molecules in our body – proteins, ions, carbohydrates, etc.
- There are also lipids.
  - Less popular in many fields than proteins, yet all of us have them.
  - … a lot of them.

TEACHING WEEK	DATE	LECTURER	TOPIC
01	21.09	TR	Getting Started (Linux, Python, VMD)
02	05.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part I
03	12.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part II
04	19.10	DB	Ions, Molecules, and Their Interactions
05	26.10	OK	Making Drugs *in silico* – part I – Lipid Membranes
06	02.11	OK	Making Drugs *in silico* – part II – Ligands Parametrizations
07	09.11	TR	Making Drugs *in silico* – part III – Steered Molecular Dynamics & Free Energy Calculations
08	16.11	TR	Making Drugs *in silico* – part IV – Free Energy Calculations & Permeability Coefficient
09	23.11	WM	How to Beat AlphaFold – part I
10	30.11	WM	How to Beat AlphaFold – part II
11	07.12	WM	Working in a Professional Environment
12	14.12	OK	Zápočet – first session to start your project

Today program

Lecture (1st hour)

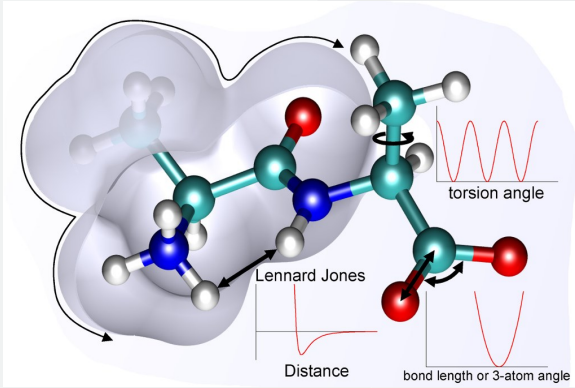
- Force fields: what, how, and when.
- Nonbonded interactions: origin and implementation.
- Periodic boundary conditions, cutoffs, and Ewald summation.

Practical session (2nd and 3rd hours)

- Preparing solvent–solute systems: ions and lipids in water.
- Solute–solute and solute–solvent interaction-driven phenomena and their analysis.

Force field: basic information

- The force field (FF, V) determines the force that an atom feels.
- The force field is “invariable” during a simulation.
- The force fields are usually additive/modular.
- The force field is the collection of functions and its parameters.
- There are many force fields for biosimulations.
  - Different in the functional forms, which reflect certain philosophy.
  - The same functional form but simply different parameters.
- Is there the ultimate and the best of the best force field?
  - No.

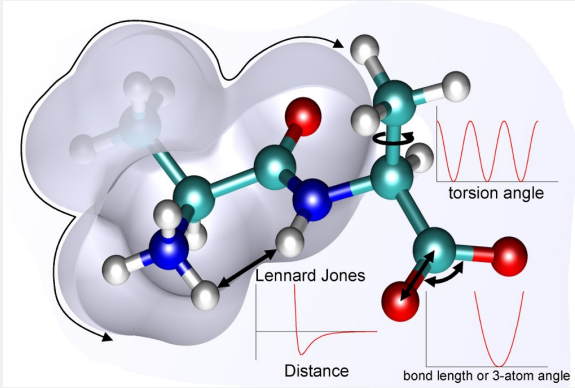


taken from Wikipedia

$$V(r_1, r_2, \ldots, r_N) = V_{bond} + V_{angle} + V_{dihedral} + V_{Coulomb} + V_{vdW}$$

Force field: basic information

- Force field – derived by fitting to experimental and/or quantum chemical data.
- Force fields are called “empirical potentials”.
  - They are not “purely” empirical, because quantum chemical data are also used for their derivation.
- There is a lot of “magic” in the force fields too.
  - It is not necessarily bad.
  - Compromises should be made.



taken from Wikipedia

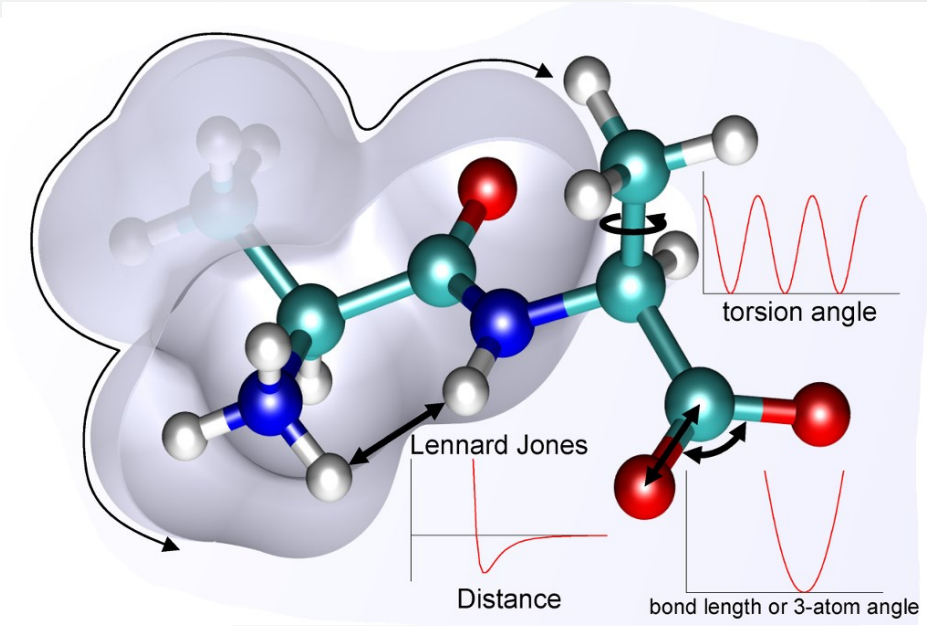
$$V(r_1, r_2, \ldots, r_N) = V_{bond} + V_{angle} + V_{dihedral} + V_{Coulomb} + V_{vdW}$$

Force field: basic information

- Typical biological simulation model does not have chemical reactions.

- Force field contains:
  - Intramolecular interactions (chemical, bonding):
    - Interactions of atoms within one molecule.
    - Bonded interactions: bonds, angles, dihedrals.

  - Intermolecular interactions (physical, non-bonding):
    - Interactions of atoms belonging to different molecules.
    - Nonbonded interactions: generally speaking, electrostatic and van der Waals (many details though).

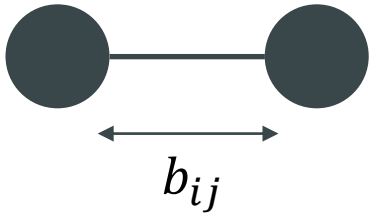


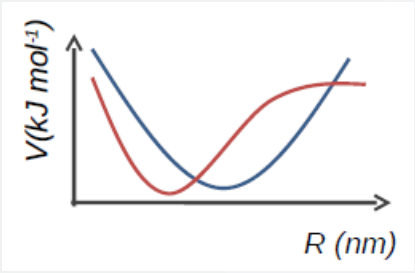
taken from Wikipedia

$$V(r_1, r_2, \ldots, r_N) = V_{bonded} + V_{nonbonded}$$

Bonded parameters: V_{bond}

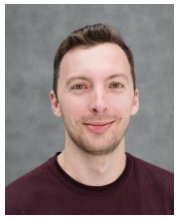
- Define “bonds” (no surprise), their equilibrium length, and stiffness.
- Typically defined by either:
  - Harmonic bond potential:
    - The most popular one in bio force fields.
    - Less accurate when studying events involve significant bond stretching.
    - Computationally cheaper.
    - Still good.
  - Morse bonds potential:
    - Not currently popular.
    - More accurate when out of equilibrium.
    - More computationally expensive.
    - Good.
- Quite often, the bonds are constrained.
  - It allows larger time step.
  - Week 09.





$$V_{bond-h}(r_{ij}) = \frac{1}{2} k_{ij}^{(b)} \left(r_{ij} - b_{ij}\right)^2$$

$$V_{bond-m}(r_{ij}) = D_{ij} \left[1 - \exp\left(-\beta_{ij}\left(r_{ij} - b_{ij}\right)\right)\right]^2$$



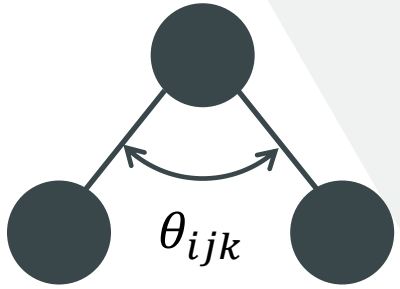
Willian Morton

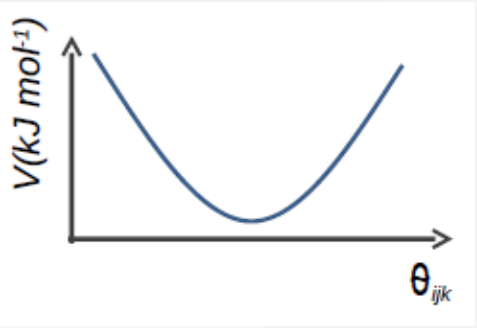
Angle parameters: V_{angle}

- Define angles, their equilibrium value, and stiffness.

- Typically defined by either:
  - Harmonic angle potential.
    - The most popular one in bio force fields.
    - Good.
  - Cosine-based potential.
  - Restricted bending potential.
  - Urey–Bradley potential.

- Angles can also be constrained in solvents.
  - Water (almost always).
  - The model being rigid implies that the geometry is fixed (both bonds and angles).

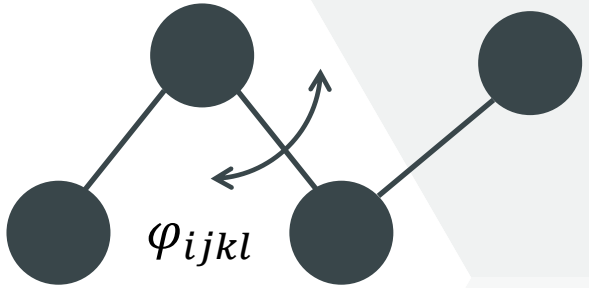


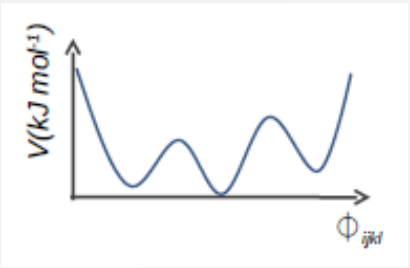


$$V_{angle-h}(\theta_{ijk}) = \frac{1}{2} k_{ijk}^{(\theta)} \left(\theta_{ijk} - \theta_{ijk}^{(0)}\right)^2$$

Dihedral (proper) parameters: $V_{dihedral}$

- Define dihedrals – torsional angles formed by four consecutively bonded atoms.
  - Angle between two consecutive planes (one defined by first three atoms, second defined by last three atoms).





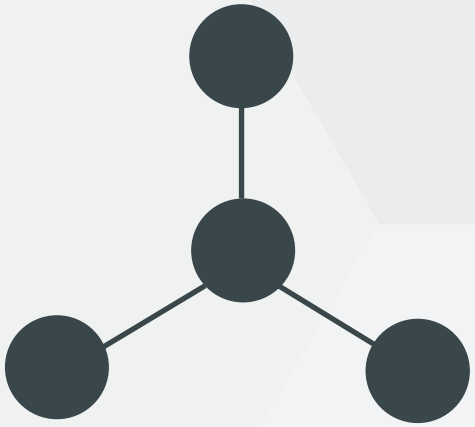
- Functional forms:
  - Periodic (Fourier, proper) (formula to the right).
  - Ryckaert–Bellemans potential.

$$V_{dihedral}(\varphi_{ijkl}) = k_{ijkl}^{(\varphi)}(1 + \cos(n\varphi_{ijkl} - \delta))$$

- Dihedrals are essential in conformation and dynamics of biomolecules.
  - Backbone rotations, nucleic acids, etc.

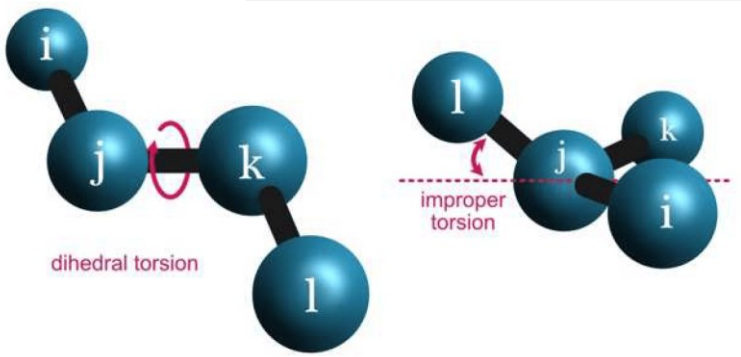
  - n – periodicity/multiplicity of the dihedral term (how many minima)
  - δ – phase shift (aka angular offset)
    - Typically, 0 for odd n and 180 for even n

Dihedral (improper) parameters: $V_{dihedral}$



- Another possible dihedral (improper one) – one atom connected to three others.

- Can be defined either as:
  - Harmonic potential (bottom–right).
    - Common choice.
    - Good.
  - Or using the same potential as for (proper) dihedral.



Fernandes et al. *Molecules* 2022, 27, 5524

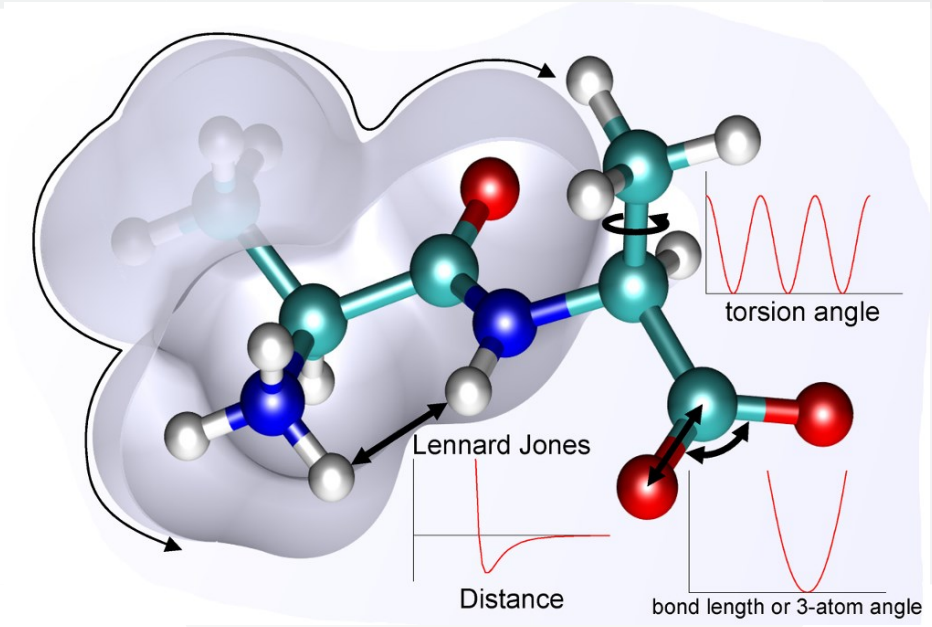
- Improper dihedrals (impropers) are also essential in conformation and dynamics of biomolecules.
  - Planar groups, chirality, etc.

$$V_{dihedral-imp}(\varphi_{ijkl}) = \frac{1}{2} k_{ijkl}^{(\varphi)} \left(\varphi_{ijkl} - \varphi_{ijkl}^{(0)}\right)^2$$

Nonbonded interactions: $V_{nonbonded}$

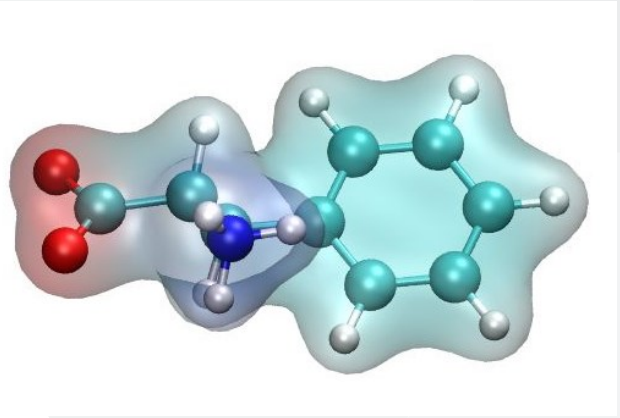
- Electrostatic interactions:
  - Charge–charge interactions (two ions).
  - Charge–dipole interactions (ion and water molecule).
  - Charge–induced dipole interactions (ion and non-polar atom/molecule).
  - Charge–(higher multipoles).

- van der Waals (vdW) interactions (all but the above, both attractive and repulsive, between uncharged particles):
  - Dipole–dipole interactions (water with another water).
  - Dipole–induced dipole, or Debye force (water with non-polar atom/molecule).
  - Induced dipole–induced dipole, or London dispersion force (two non-polar atoms/molecules).
  - Pauli repulsion (repulsive interaction, more details later).

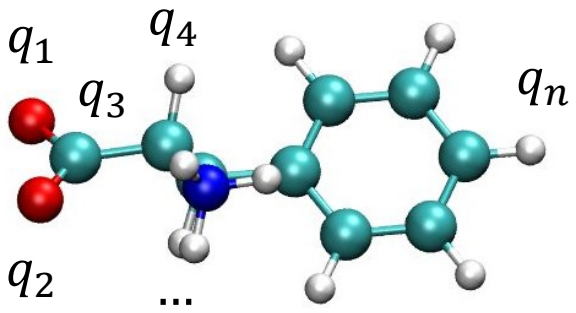


taken from Wikipedia

Electrostatics is the core



- All interactions at molecular level have their origin in electrostatics.
- Biosimulations typical use nonpolarizable force fields.
  - Electrostatic interactions are approximated as charge–charge interactions.
- Electronic clouds represented as point charges.
  - Derived from the approximate electrostatic potential.
  - Electrostatic potential is calculated from quantum mechanics methods exist (ESP, RESP, etc.)
  - Atomic partial charges are not experimental observables, definition is arbitrary.
    - Week 06.



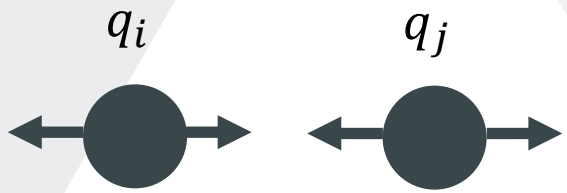


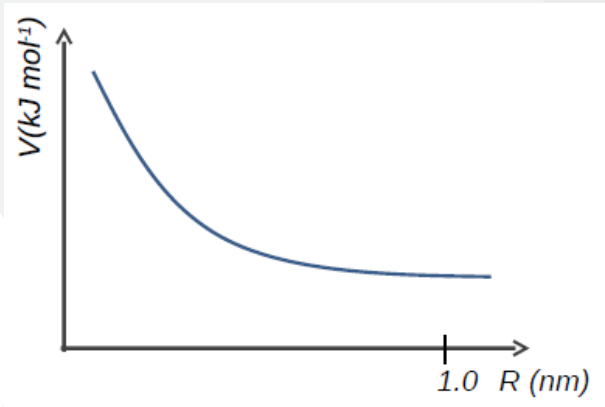
Ondřej Kroutil

Charge–charge interactions: $V_{Coulomb}$

- Described by Coulomb law:

$$V_{Coulomb}(r_{ij}) = \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$





- Charge–charge interactions drive multiple processes.
  - Dominant interactions.
  - Both short-range and long-range.

- Other contributors (like charge–dipole) are smaller.
  - Do not appear explicitly in nonpolarizable force fields.
  - Appear explicitly in polarizable force field (outside the scope of the course).

van der Waals (V_{vdW}) attractive interactions

Interaction	Dependence on distance	Typical energy (kJ/mol)
charge–charge	$1/r$	~250
charge–dipole	$1/r^2$	~15
dipole–dipole	$1/r^3$	~2
London force	$1/r^6$	~2
hydrogen bond		~20

based on Atkins' Physical Chemistry

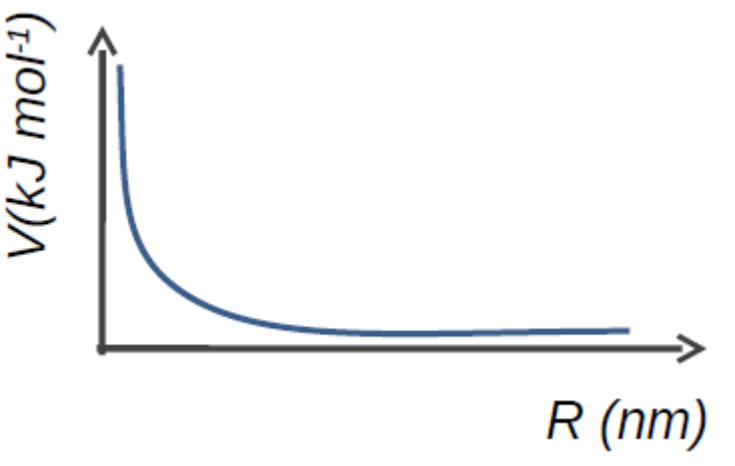


- Attractive vdW interactions can be approximated as:
  - $C_{ij}^{(6)}$ – related to the strength of attractive interactions

$$V_{vdW-att}(r_{ij}) = -\frac{C_{ij}^{(6)}}{r_{ij}^6}$$

van der Waals (V_{vdW}) repulsive interactions

- Short-range vdW interactions are strongly repulsive:
  - Pauli repulsion.
  - Overlap of electronic densities.
    - Two electrons cannot occupy the same quantum state at the same time.
  - Decays to zero at long distances.



- Originates from quantum mechanical exchange energy between fermions.

$$V_{vdW-rep}(r_{ij}) = \frac{C_{ij}^{(12)}}{r_{ij}^{12}}$$

- Can also be approximated (formula to the right)
  - $C_{ij}^{(12)}$ – related to the strength of repulsive interactions



van der Waals as Lennard–Jones: $V_{vdW} = V_{LJ}$

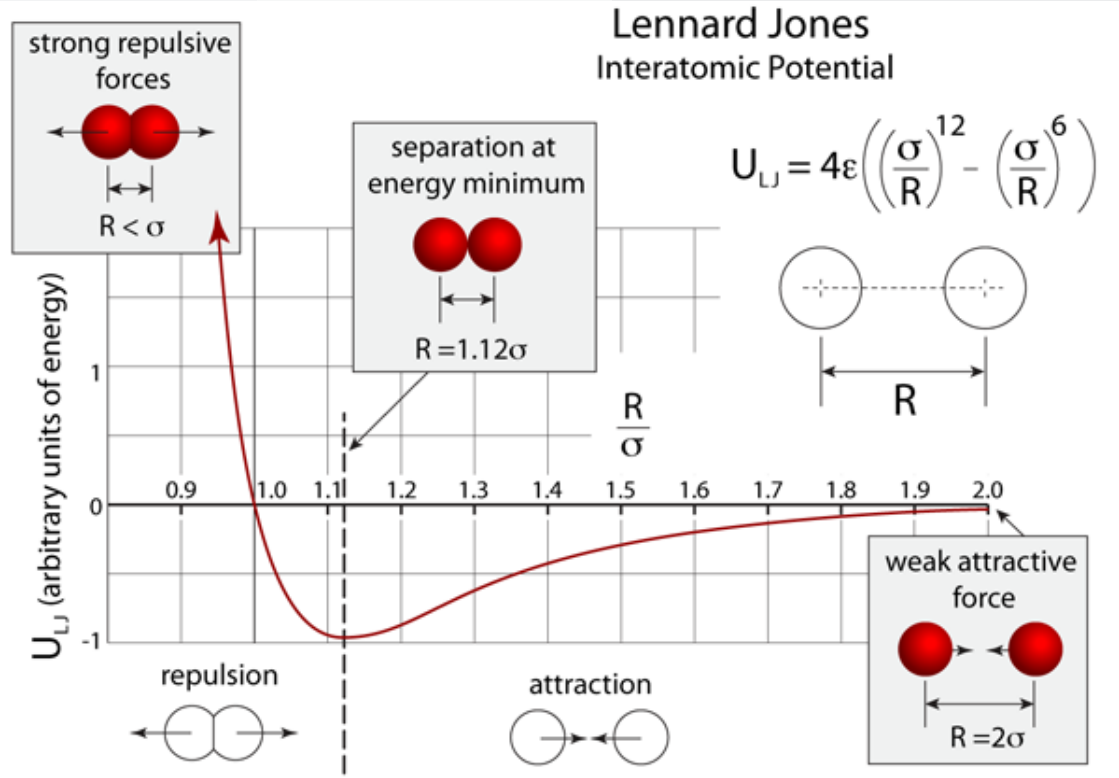
- Attractive and repulsive components constitute vdW interactions
  - C_{12} – related to the strength of repulsive interactions
  - C_6 – related to the strength of attractive interactions

$$V_{vdW}(r_{ij}) = \frac{C_{ij}^{(12)}}{r_{ij}^{12}} - \frac{C_{ij}^{(6)}}{r_{ij}^{6}}$$

- Can be approximated as Lennard–Jones potential:
  - σ_{ij} – depth of the potential that indicate the strength of the attraction
  - ϵ_{ij} – finite distance at which the potential is zero

$$V_{LJ}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

Lennard–Jones potential in a nutshell



http://atomsinmotion.com/book/chapter5/md

- Atoms always try to minimize their potential energy.
- At the lowest temperatures (lowest kinetic energy), they are falling to the bottom of the potential curve.
- Deviation to the left – atoms start to repel.
- Deviation to the right – atoms start to attract each other.

Lennard–Jones potential: combining/mixing rules

- σ and ϵ are typically given as atomic properties.
  - σ_{ii} and ϵ_{ii} are estimated.
  - Force field usually have only those, not σ_{ij} and ϵ_{ij}.
  - σ_{ij} and ϵ_{ij} can be then tuned explicitly to reproduce experimental properties.

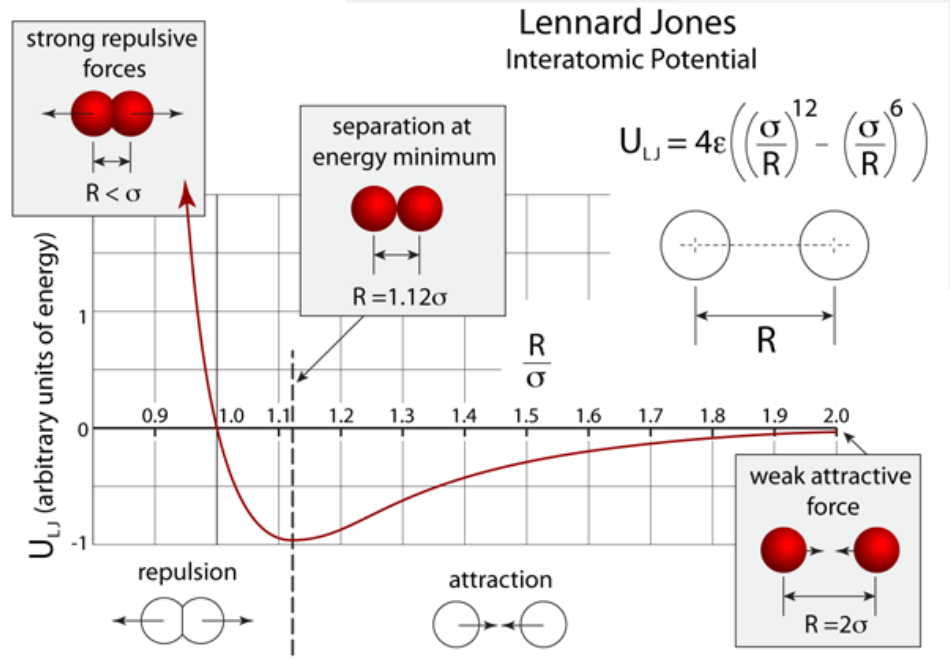
$$V_{LJ}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

- Otherwise, σ_{ij} and ϵ_{ij} are derived from combining/mixing rules.
  - Geometric rules:

$$\sigma_{ij} = \left(\sigma_{ii}\sigma_{jj}\right)^{1/2} \qquad \epsilon_{ij} = \left(\epsilon_{ii}\epsilon_{jj}\right)^{1/2}$$

  - Lorentz–Berthelot rules:

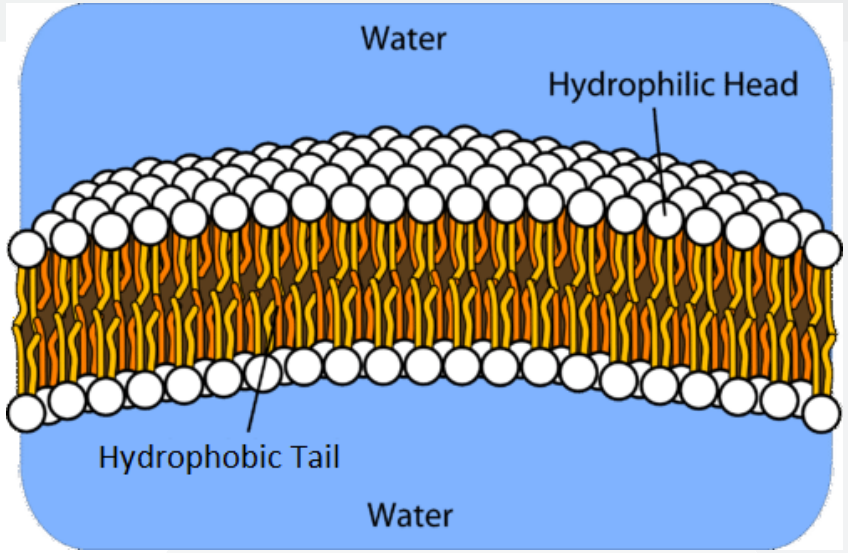
$$\sigma_{ij} = \frac{1}{2}\left(\sigma_{ii} + \sigma_{jj}\right) \qquad \epsilon_{ij} = \left(\epsilon_{ii}\epsilon_{jj}\right)^{1/2}$$



http://atomsinmotion.com/book/chapter5/md

Hydrophilic and hydrophobic interactions

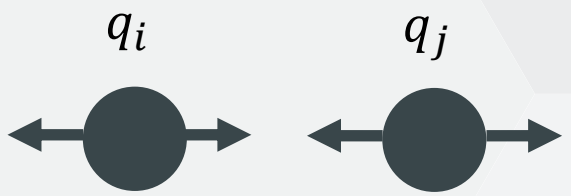
- Molecules are not “stupid”.
- Non-polar molecules tend to interact with each other in a polar solvent (e.g., water).
- Hydrophobicity/hydrophilicity results from electrostatic and van der Waals interactions.
  - Hydrophilicity: polar molecules (molecules with charged groups) like to interact with water,
  - Hydrophobicity: non-polar molecules (molecules without charged molecules) do not prefer to interact with water
- Hydrophobic interactions is a disputable term.
  - Analogy to light and darkness.



Schematical representation of a membrane

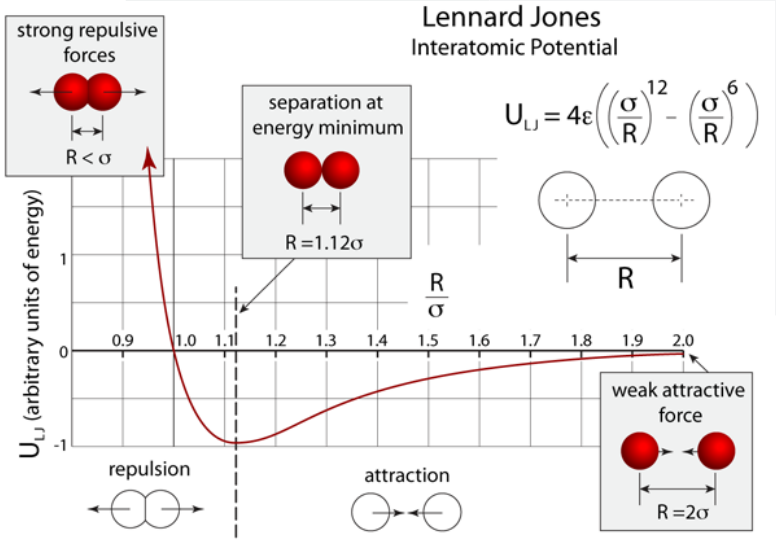
https://chem.libretexts.org/

Nonbonded interactions: $V_{nonbonded}$



- Electrostatic interactions:
  - **Charge–charge interactions (two ions).**
  - ~~Charge–dipole interactions (ion and water molecule).~~
  - ~~Charge–induced dipole interactions (ion and non-polar atom/molecule).~~
  - ~~Charge–(higher multipoles).~~

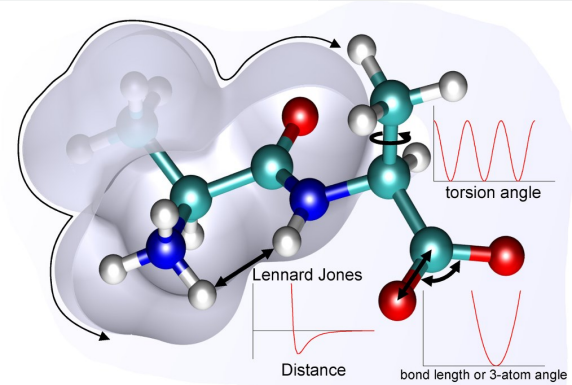
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  - Dipole–induced dipole, or Debye force (water with non-polar atom/molecule).
  - Induced dipole–induced dipole, or London dispersion force (two non-polar atoms/molecules).
    - **Simplified to attractive term.**
  - **Pauli repulsion (repulsive interaction).**



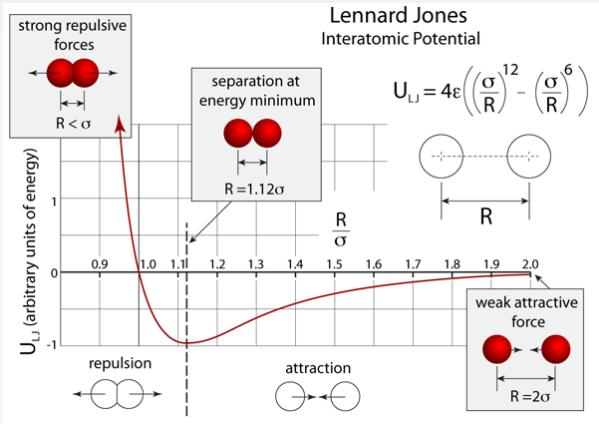
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Force fields: remarks

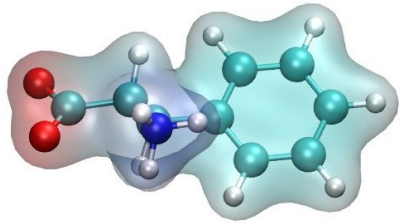
- Based on experiments, quantum chemistry data, or until it works.
- Embodied in interaction models, which are practical and fast.
- Quite often system-specific, sometimes – field-specific
  - There are force field designed for specific type of molecules (e.g., proteins).
  - There are more generic force fields, like those for ions, biomolecules, solid particles, etc.
- Today we discussed nonpolarizable force fields.
  - Explicitly/implicitly polarizable force fields are another level of complexity.
    - Consider not only charge–charge interactions.
    - Supposed to be better but usually not.
- Not suitable/problematic for certain phenomena (e.g., chemical reactivity, excited states) or conditions (low temperature, high pressure, unless intentionally designed).
  - Other devil details.



taken from Wikipedia

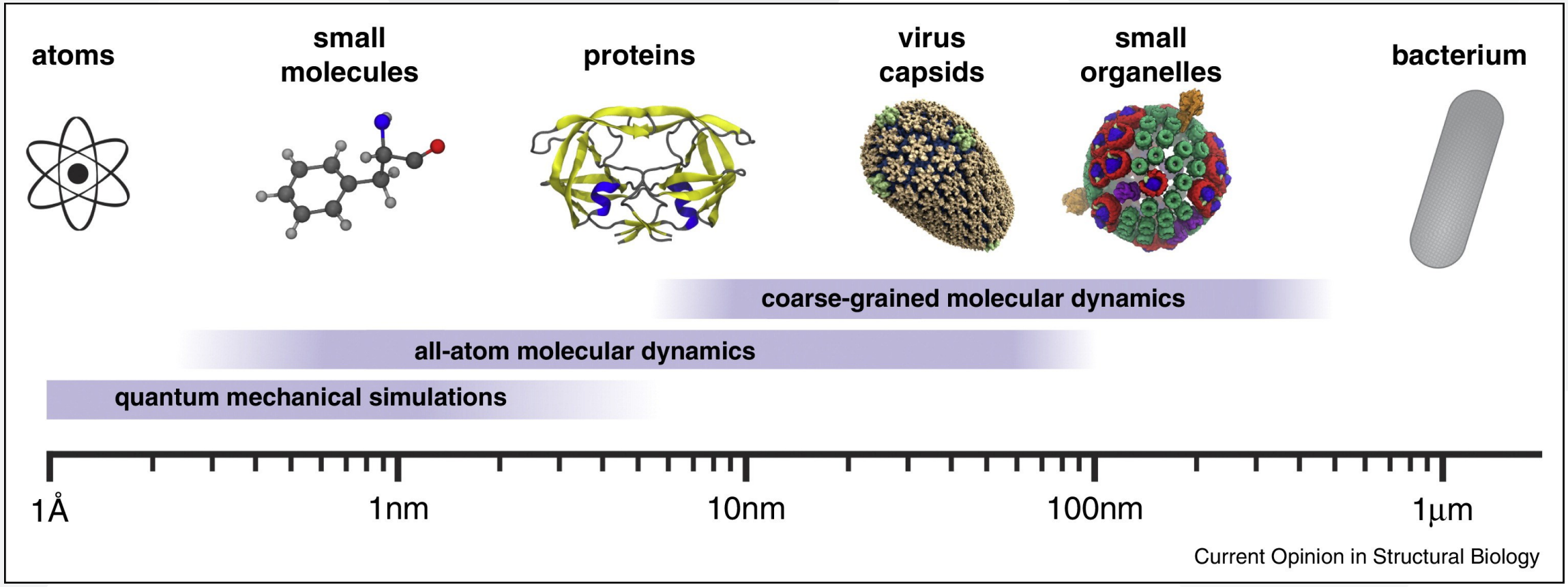


http://atomsinmotion.com/book/chapter5/md



- **FORCE FIELDS ARE THE KEY TO MOLECULES DYNAMICS.**
  - **Knowledge comes with experience and monitoring literature.**

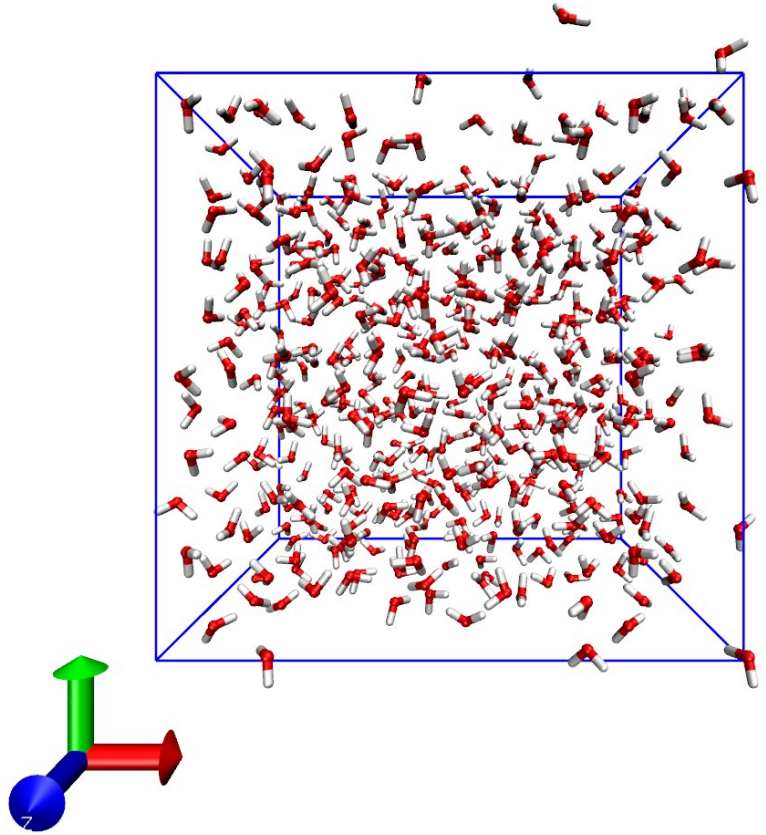
Size of biomolecules and biosystems



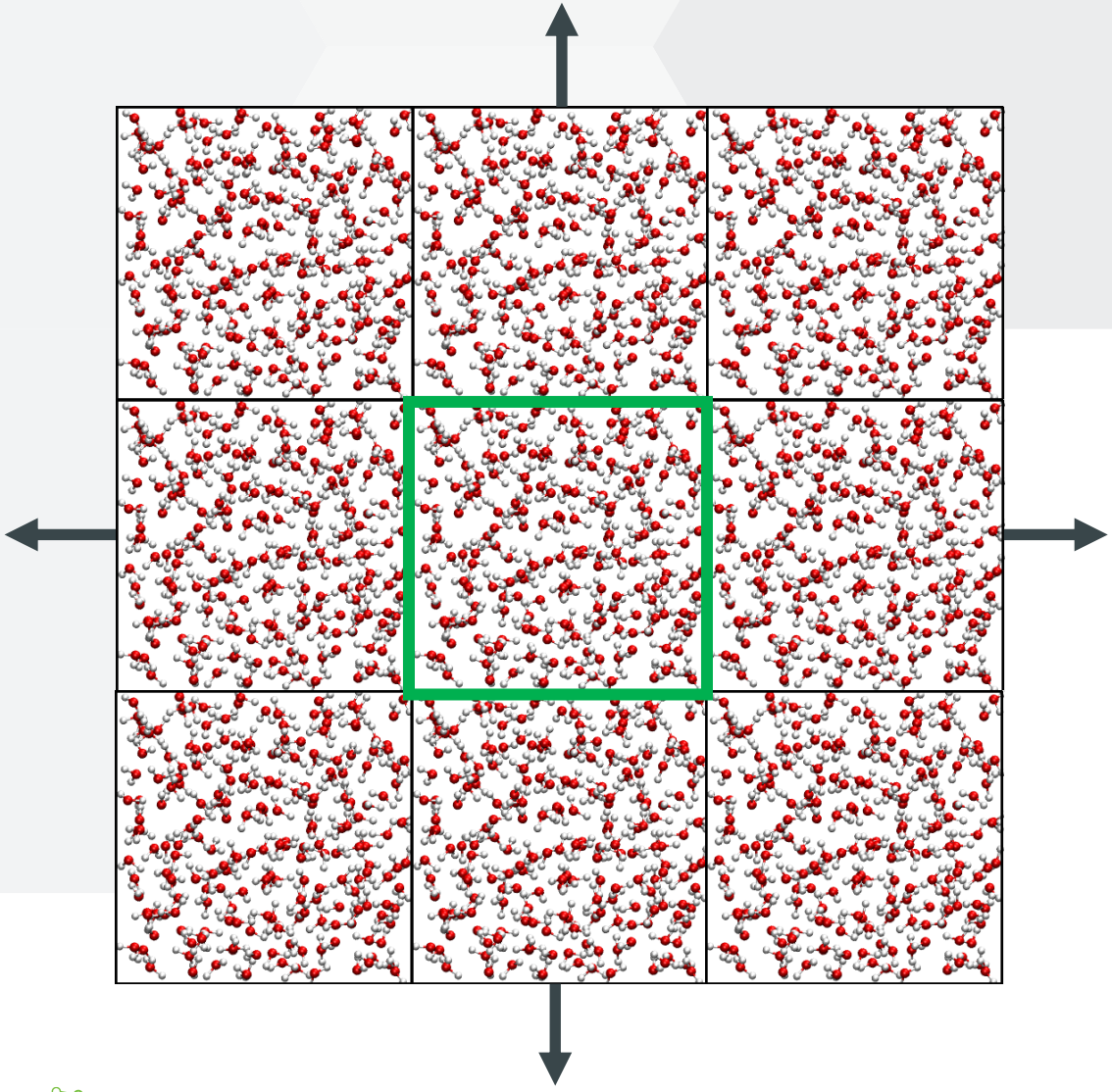
Perilla et al. *Curr. Opin. Struct. Biol.* 2015, 31, 64–74.

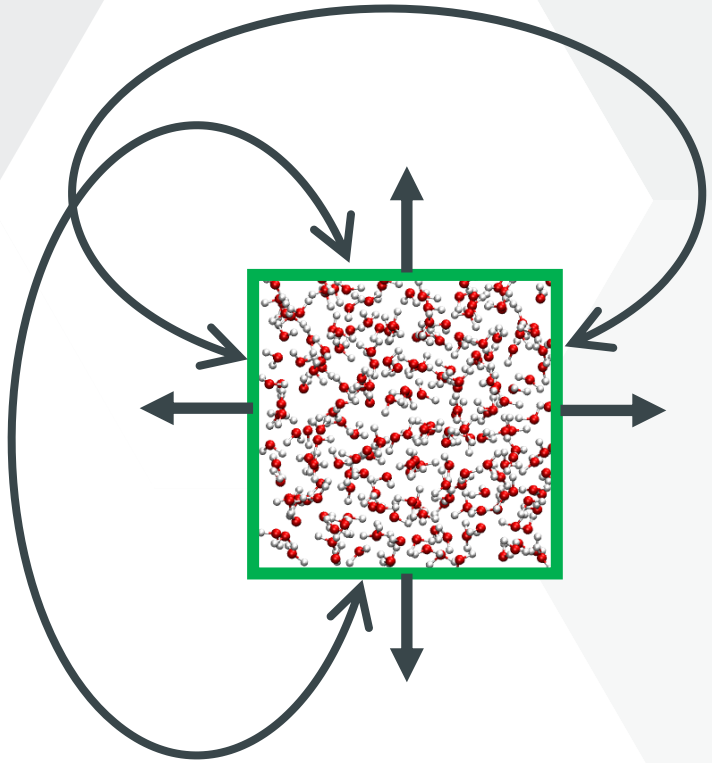
Periodic Boundary Conditions (PBC)

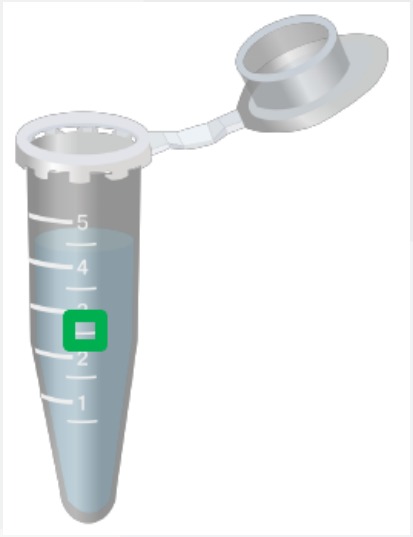
- Simulations boxes are finite in size.
  - They represent a part of physical space.
  - … and they are clearly not micrometers long.
- The simulated systems are:
  - Typically, 100,000–1,000,000 atoms.
  - Have a cubic box size up to ~10–20 nm.
- Suitable *per se* for isolated small molecules and small molecular clusters.
- Not suitable for condensed phases
  - Proteins in water, cellular membranes, etc.
- Solution: **periodic boundary conditions (PBC)**



Periodic Boundary Conditions (PBC)



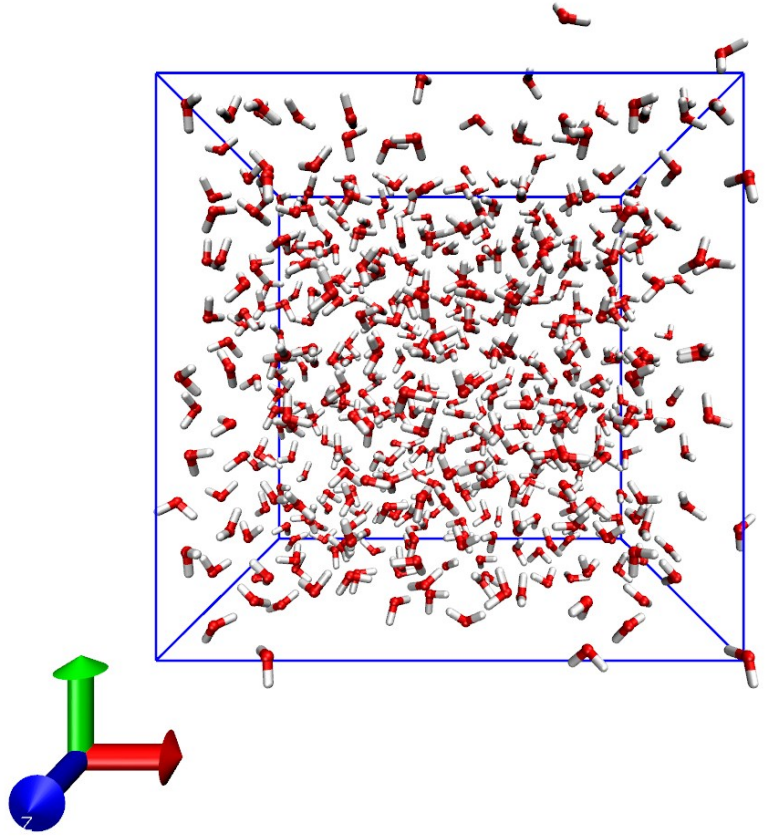




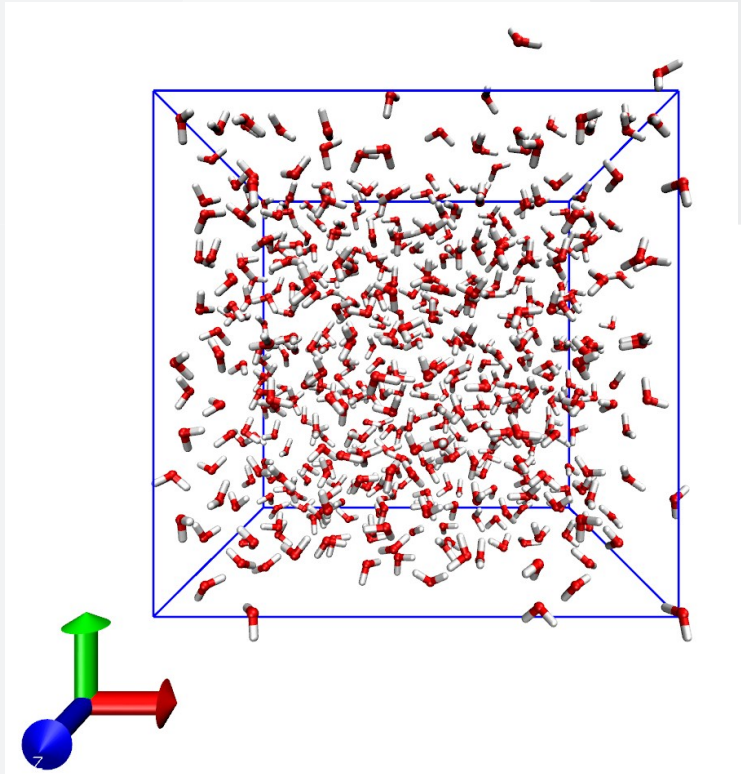
taken from Wikipedia

Periodic Boundary Conditions (PBC)

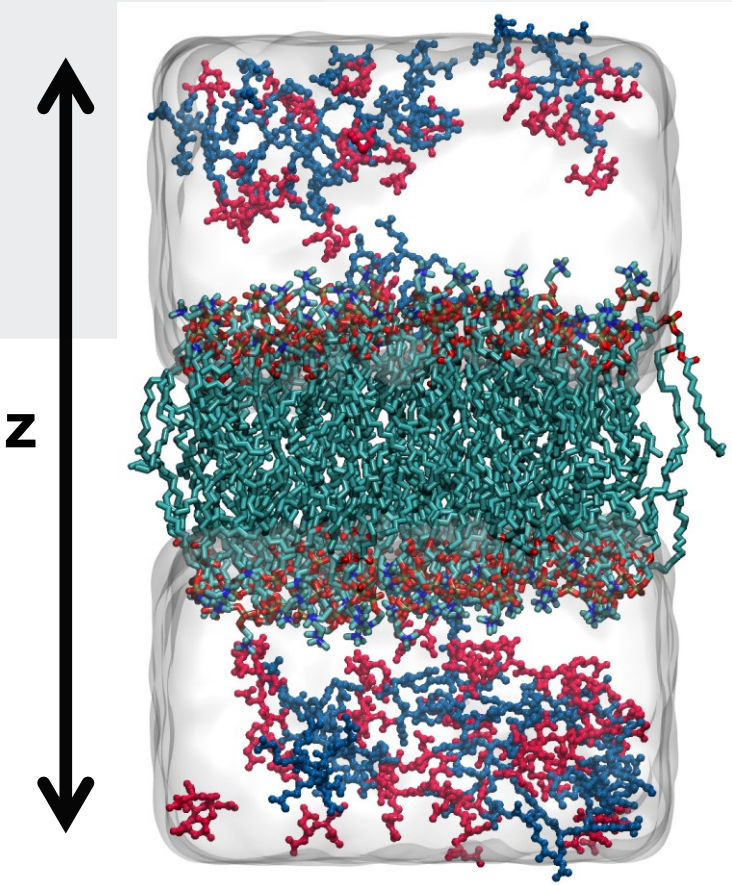
- Finite system becomes infinite.
- Pros:
  - Artificial edges eliminated.
  - Almost no increase in computational cost.
  - Particularly good for inherently periodic systems, e.g., crystals.
  - Suitable for liquid phases too.
- Cons:
  - Periodicity introduces certain artifacts itself.
    - Remember previous lesson about self-diffusion and dependence on the box size?
  - Might not be suitable for phenomena that occur at lengths greater than the cell size.
  - Might not be suitable for studying local defects and non-idealities.



PBC: specific examples

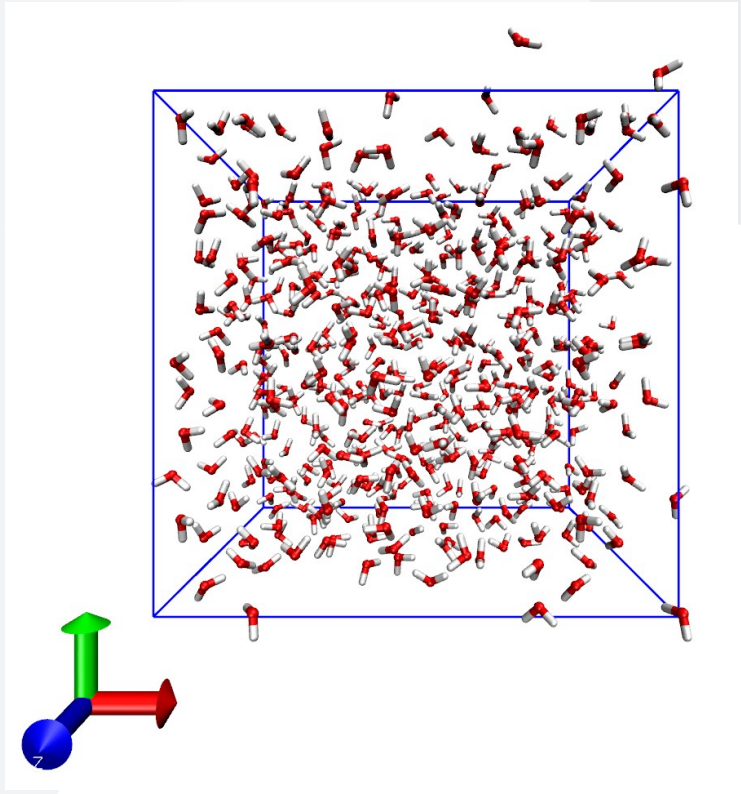


Box with water, simple to think about

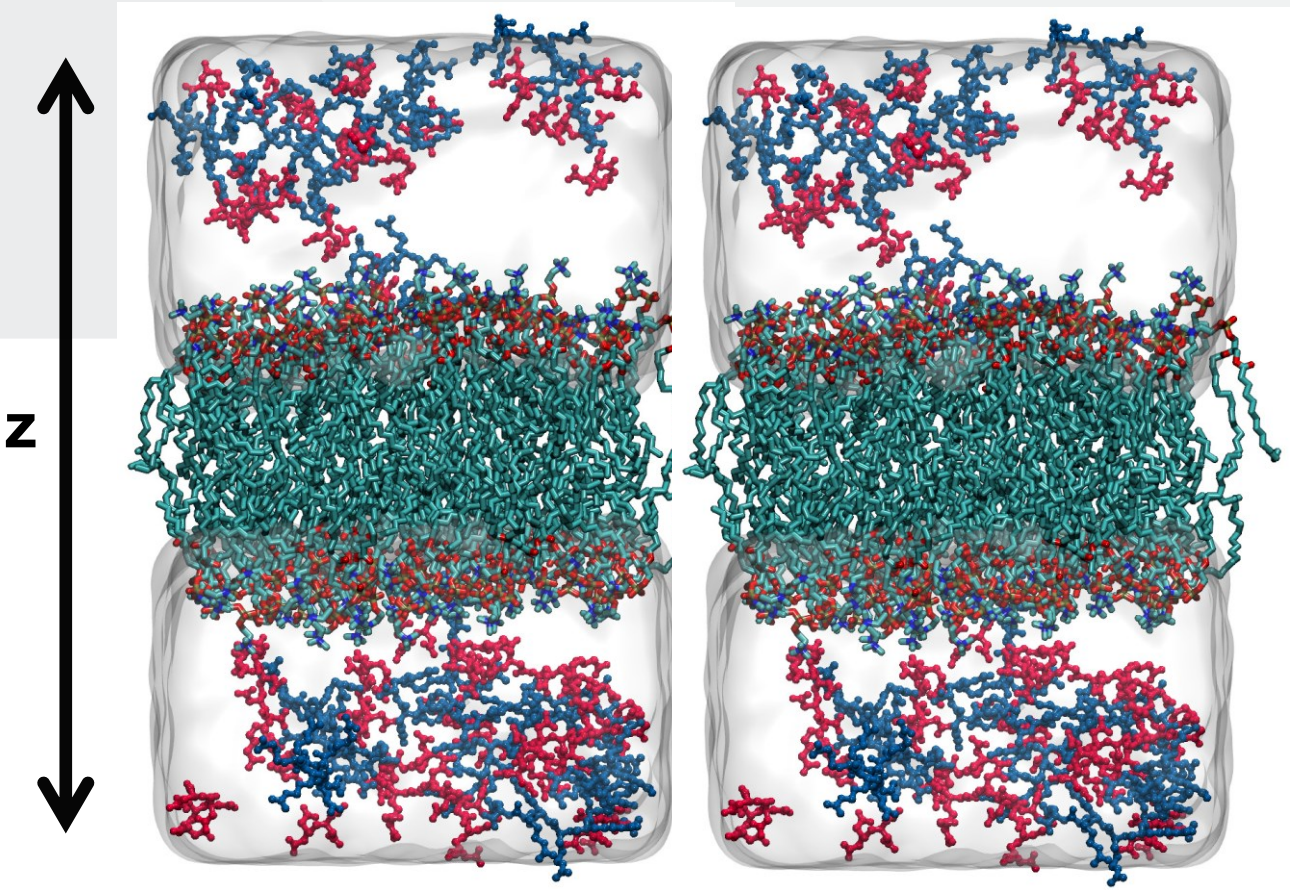


Lipid membranes, suitable,
but more cautious;
the same applies to any interfacial system

PBC: specific examples



Box with water, simple to think about

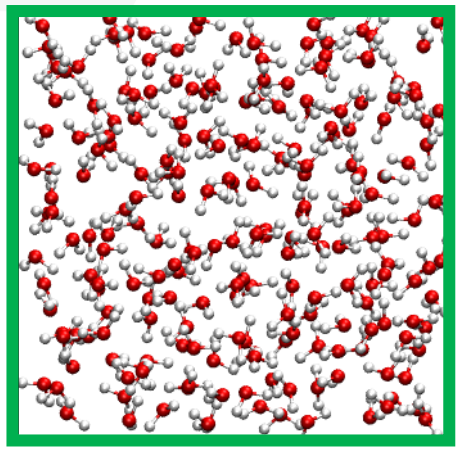


Lipid membranes, suitable,
but more cautious;
the same applies to any interfacial system

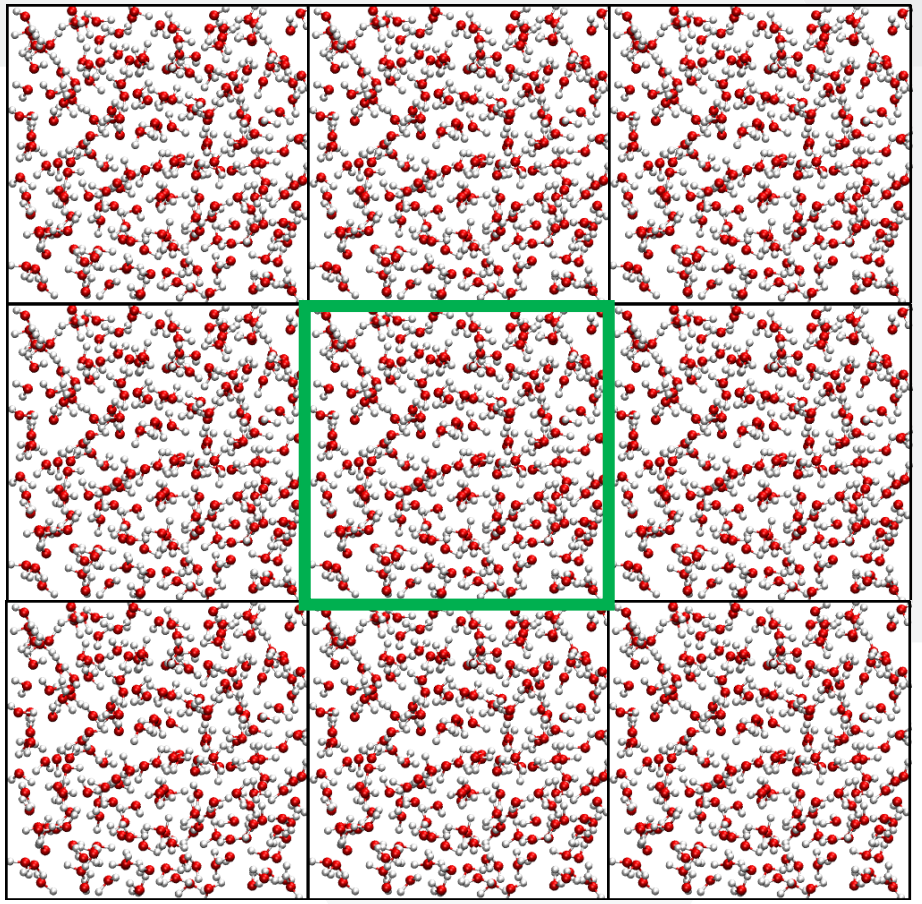
PBC: details to consider

- Box still should be large enough.
  - Common sense.
  - Interactions with itself should be prevented unless intended.
  - Interactions within only half box size should be considered.
  - Technical details of simulations should be adjusted accordingly (next slide).

$L/2$ – half box size



L – box size



Cutoffs: what is that?

- A predefined distance beyond which interactions are ignored or treated approximately.
- For van der Waals interactions, beyond certain distances interactions are negligible.
  - Can be ignored.
  - Saves computational time.
- For electrostatics, cannot be ignored.
  - Long-range nature of interactions.
  - Approximated.
    - For instance, using Particle Mesh Ewald Method.
    - Saves computational time.
- In both cases, interactions are usually not suddenly truncated.
  - To not cause artifacts.

if r_{ij} < cutoff:

$$V_{LJ}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

$$V_{Coulomb}(r_{ij}) = \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$

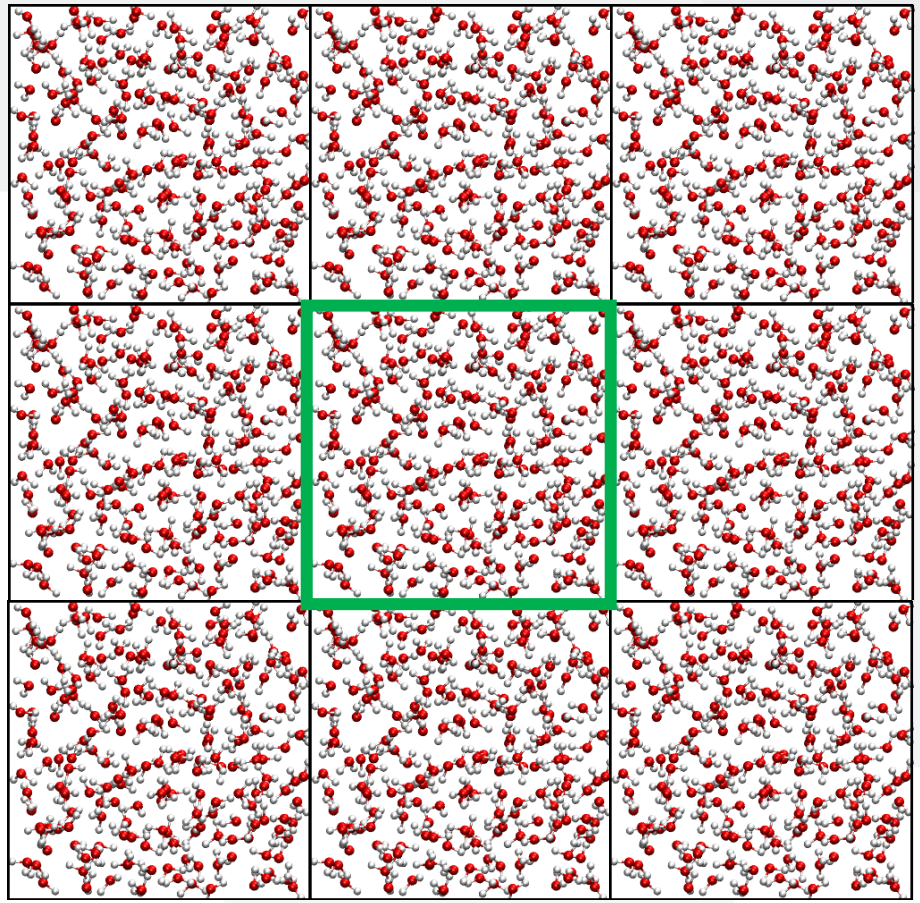
if r_{ij} > cutoff:

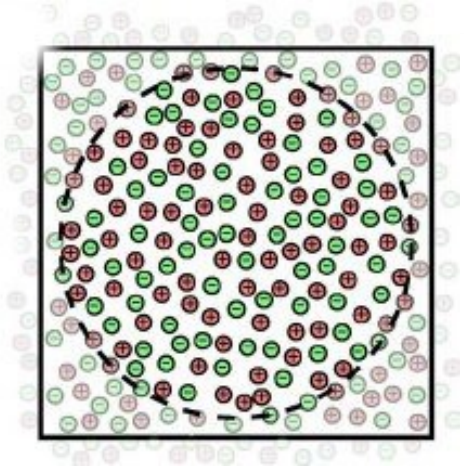
$$V_{LJ}(r_{ij}) = 0$$

$$V_{Coulomb}(r_{ij}) = \text{approximated}$$

Particle Mesh Ewald (PME)

- A technique to efficiently compute long-range electrostatic interactions.
- PME divides electrostatic interactions into two components:
  - Real space (Coulomb law as we discussed).
  - Reciprocal space (mostly from periodic images).
    - Fast Fourier transformation.
    - Interpolation (particles charges put onto a mesh to compute the potential).
- Conditions:
  - System is in PBC.
  - Total charge of the system is zero.
- Advantages:
  - Highly accurate.
  - Flexible.

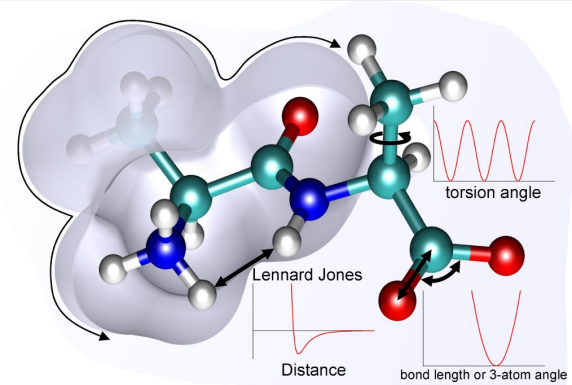




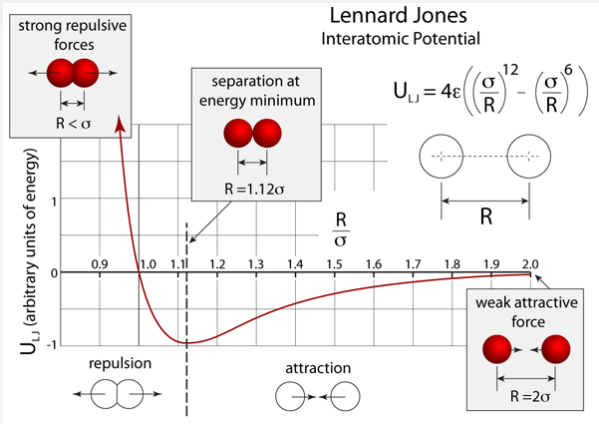
Fennell and Gezelter, *J. Chem. Phys.* 2006, 124, 234104.

Force fields: more remarks

- Cutoff values are force-field specific.
  - If the force field was designed using specific cutoff, PME method or alternative, then this protocol should be used in all other simulations with this force field.
- Yes, simulations have protocols like wet-lab experiments.
- The same applies to things like time step, type of thermostat and barostat, etc.
  - More about thermostats and barostats – week 05.



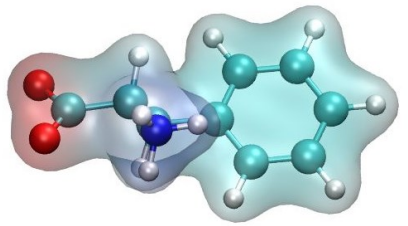
taken from Wikipedia



http://atomsinmotion.com/book/chapter5/md



Ondřej Kroutil



Break!

Get ready for practical session!